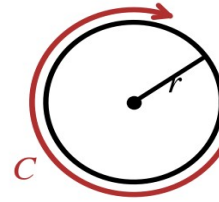


Speak Like A Mathematician:

circumference:

arc length:

radian:



Circumference and Arc Length

Circumference: $C = 2\pi r = \pi$ _____ .

Arc length: $s = (\text{arc measure} / \text{_____}) \cdot 2\pi r$ — an arc is a **FRACTION** of the circumference.

Radians: $180^\circ = \text{_____}$ radians, so multiply degrees by $\pi/180$ to convert.

Example 1 - Circumference Both Ways

- a) A circle has radius 7. Find its circumference exactly, then to the nearest tenth.
- b) A circle has diameter 18. Find C.
- c) A circle has circumference 20π . Find its radius.

Example 2 - Arc Length

In the top figure, the central angle is 135° and the radius is 8.

- a) Find the arc length s exactly, then to the nearest tenth.

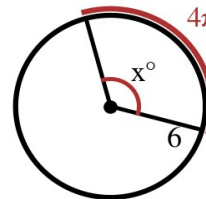
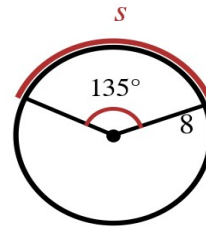
Example 3 - Another Arc

- a) Find the length of a 60° arc in a circle of radius 9.

Example 4 - Working Backward

In the bottom figure, an arc of length 4π sits in a circle of radius 6.

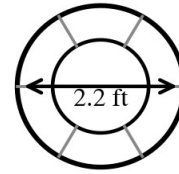
- a) Find x , the measure of the central angle.



Example 5 - How Far Does a Tire Roll?

A bicycle tire has diameter 2.2 ft.

- a) How far does the bike travel in ONE full revolution of the tire?
- b) About how many revolutions does the tire make in 500 ft?



Example 6 - Radians

- a) Convert 120° to radians (exact).
- b) Convert $\pi/6$ radians to degrees.
- c) Why are radians natural for arc length?